



Evaluating NeRFs for 3D Plant Geometry Reconstruction in Field Conditions

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Motivation

3D Plant Geometry's Role:

- Essential for plant growth and physiological studies.

Current Methodology Limitations:

- Fails to capture intricate details and are costly.

Why NeRF?

- 3D from 2D images, without specialized equipment.

Objective:

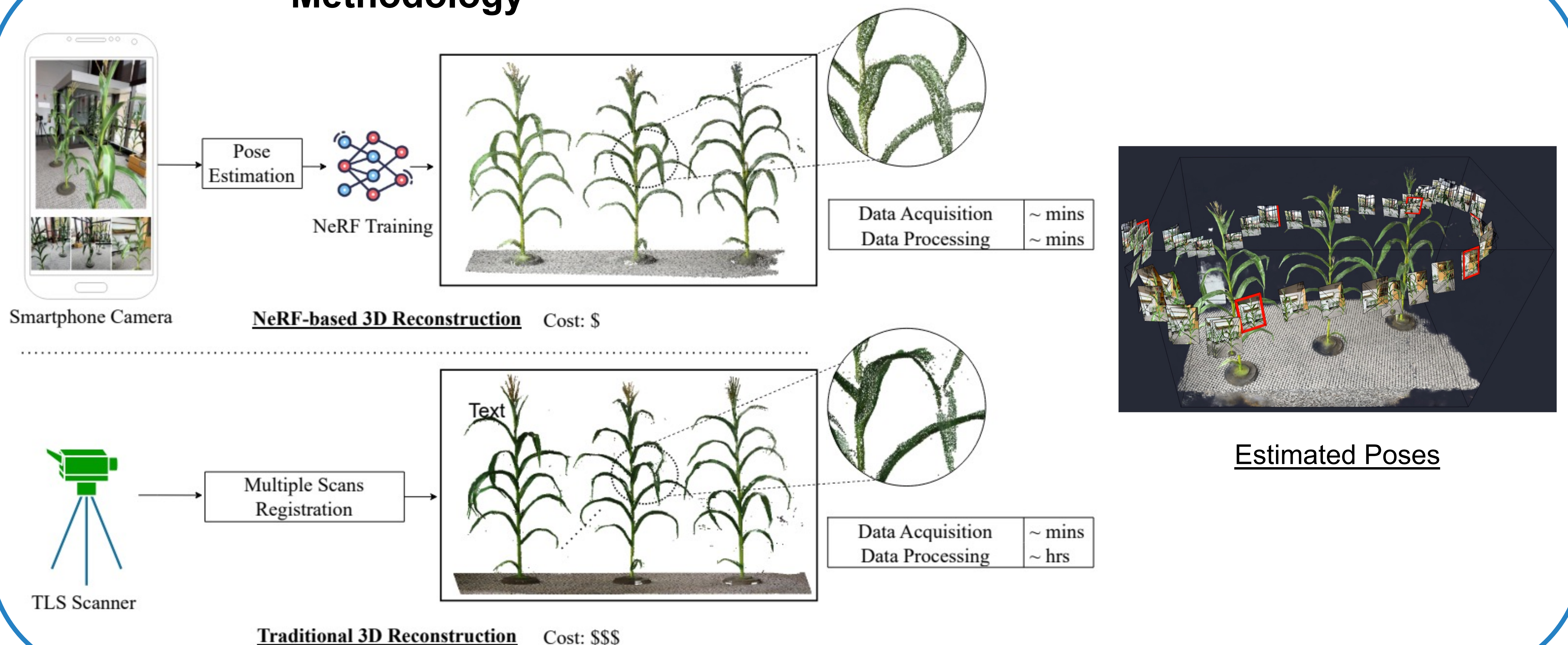
- Evaluate NeRFs for 3D plant reconstruction.

Results

#	Model	Precision ↑	Recall ↑	F1 ↑	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
I	Instant-NGP	24.66	90.62	38.77	23.41	0.81	0.17	756
	TensoRF	9.58	43.34	15.69	14.69	0.55	0.66	1973
	NeRFacto	73.57	94.72	82.81	22.24	0.73	0.12	1938
II	Instant-NGP	23.45	58.57	33.49	19.08	0.64	0.31	1886
	TensoRF	20.5	55.34	29.91	15.54	0.42	0.56	2607
	NeRFacto	64.47	76.8	70.1	18.93	0.64	0.25	1226
III	Instant-NGP	15.06	59.55	24.04	18.54	0.47	0.4	1466
	TensoRF	40.95	75.62	53.13	17.32	0.39	0.55	1965
	NeRFacto	68.29	82.32	74.65	16.7	0.32	0.34	1499

- Leveraged LPIPS to save the computation time by upto 61.1% at cost of 7.4% performance loss.

Methodology



Contributions

- Comprehensive dataset across various plant scenarios.
- Evaluation of state-of-the-art NeRF techniques.
- Early stopping algorithm for efficient training.

Conclusion

- Supports non-intrusive agricultural technologies and sets a baseline.
- Early stopping algorithm improve efficiency.



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